Solution:

Let the x-axis be the time in seconds the y-axis be the distance in metres. Let the engine be at the origin O. Therefore the length of the train 560 m is the negative y-intercept.

The uniform speed 12.5 m/s is the slope of the motion of the train. $\left(\text{speed} = \frac{\text{distance}}{\text{time}}\right)$

Since we are given slope and y-intercept, the equation of the line is

$$y = mx + b \tag{6.13}$$

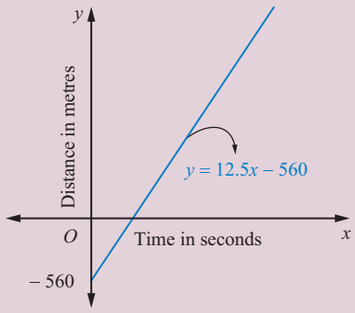


Figure 6.27

(i) The equation of the motion of the train,

$$\begin{aligned} \text{when } m &= 12.5 \text{ and } b = -560, \\ \text{is } y &= 12.5x - 560 \end{aligned}$$

(ii) When the engine touches the other side of the bridge (island)

$$\begin{aligned} y &= 2065 \text{ and } b = 0 \\ 2065 &= 12.5x \\ x &= 165.2 \text{ seconds.} \end{aligned}$$

(iii) When $y = 0$, the last coach cross the entry point of the bridge,

$$\begin{aligned} 0 &= 12.5x - 560 \\ x &= 44.8 \text{ seconds.} \end{aligned}$$

(iv) When $y = 2065$, the time taken for the train to cross the other end of the bridge is given by

$$\begin{aligned} 2065 &= 12.5x - 560 \\ x &= 210 \text{ seconds.} \end{aligned}$$

(One may take the tail of the train as the origin and can find the equation of the straight line. It need not be identical with the above equation, but the path traced out by the train, distance, time, etc.,will be the same. Try it.)

Example 6.11 Find the equations of the straight lines, making the y- intercept of 7 and angle between the line and the y-axis is 30°

Solution:

There are two straight lines making 30° with the y-axis.